

RVHS JC2 H2 Physics Preliminary Examinations Paper 3 Mark Scheme

Questions			Answers	Marks
1	(a)		Systematic errors are constant deviations of the readings of a physical quantity, either consistently higher, or consistently lower, than its true value.	[B1]
			Random errors are deviations of readings of a physical quantity, randomly scattered about the mean reading.	[B1]
	(b)	(i)	21.150 cm 20.980 cm Half the smallest division for read-off	[B1]
		(ii)	0.005 cm (half smallest division)	[B1]
		(iii)	$A = \pi(D/2)^2 = \pi D^2/4$ $\Delta A/A = 2\Delta D/D$ Using $\Delta D = (0.005 + 0.005) = 0.01$ $= 2(0.01)/(21.15 - 20.98) = 11.76 \sim 12\%$ (max 2 s.f.)	[M1] [A1]